DATA.ML.100 Introduction to PR and ML — *Final online test, October 11th 2024*

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Rules:

For every deadline you must upload new files and save changes. DO NOT upload the file again after the deadline. If you miss a deadline or submit a wrong answer, the test is over.

You NEED a computer with Internet access, Slack, and Python with libraries used in the course installed.

You are allowed to use Internet and all course materials.

You are NOT allowed to communicate with anyone in any form during the exam.

You MUST join the course Slack channel during the exam for announcements. Questions only privately to the instructor.

All code and text MUST be written by yourself. Copying substantial parts from any source is PLAGIARISM.

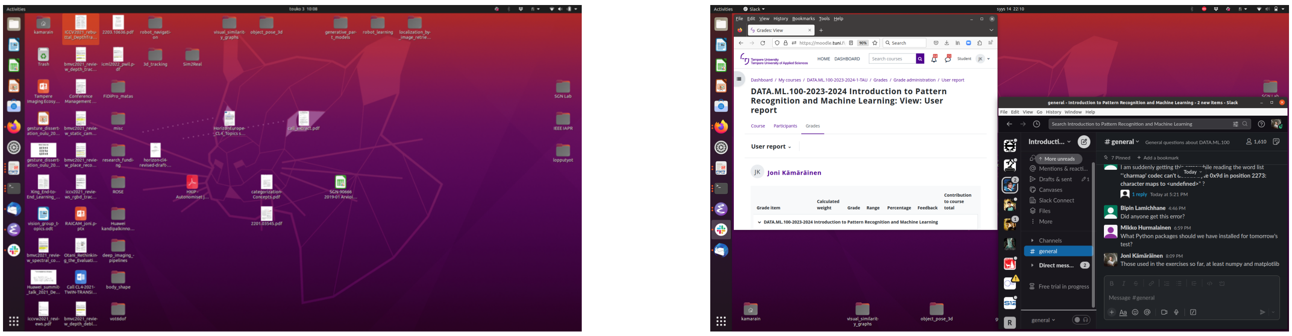
1. Basics (0pts) **[Return before 10am30]**

   Take two screenshots of your **full** desktop:

   - An empty desktop where all applications are minimized (see Fig. 1(a))
   - Open your grading page in Moodle DATA.ML.100 (your name must be visible) and the course Slack (latest message must be visible) (see Fig. 1(b))

   Submit the following items before the deadline:

   - Screenshot 1 as PNG file: `screenshot_A.png`
   - Screenshot 2 as PNG file: `screenshot_B.png`



(a) Empty desktop (b) Moodle and Slack open

Figure 1: Examples of the screenshots that should be returned

2. Fake news or true news investigation **[Multiple deadlines]**

   In 1929, the steam ship SS Kuru sank in the lake Näsijärvi. The headline of the local newspaper claims "Boat sank - only rich people were saved!". Let's analyze whether this is true or not!

   Passenger information is stored to the following file

   - `train.dat`

   The data columns are: 1. passenger number, 2. survival status (1: survived, 0: died), 3. sex (0: male, 1: female), 4. age (in years or '-1' if unknown), 5. paid trip fare (in former Finnish currency).

   (a) Basic statistics (5pts) **[Return before 10am45]**

   Read the data file (use `numpy.loadtxt()` - you may need to set `encoding='utf8'`). Write a python script to count the total number of passengers, how many survived, and how many died. The output of your python script should be

```
(dataml100) $ python code_C.py
Out of XXX passengers YYY (yy.yy%) survived and ZZZ (zz.zz%) died.
```

Submitted items:

- Full desktop screenshot of running your code and printing outputs (must include Slack with the latest comment): `screenshot_C.png`
- Python source code: `code_C.py`

(b) Fare histograms (10pts) **[Return before 11am05]**

To verify that more rich people survived than poor people, plot two separate histograms in the same picture (use different colors and alpha blending). The first histogram should be the fares paid by the survived and the second fares paid by the dead passengers. Label y and x axes correctly, and plot legend for the two histograms.

```
(dataml100) $ python code_D.py
```

Submitted items:

- Full desktop screenshot of running your code and plotting the histograms (must include Slack with the latest comment): `screenshot_D.png`
- Python source code: `code_D.py`

(c) Alternative explanation (10pts) **[Return before 11am30]**

Investigate whether other attributes than the ticket fare can explain who survived. Count how many of the male passengers survived, and how many of the female passengers survived. (you may find the functions `numpy.intersect1d()` and `numpy.setdiff1d()` useful)

```
(dataml100) $ python code_E.py
Out of XXX male passengers YYY survived and ZZZ died
   (surveillance rate yy.yy%)
Out of XXX female passengers YYY survived and ZZZ died
   (surveillance rate yy.yy%)
```

Submitted items:

- Full desktop screenshot of running your code and printing the results (must include Slack with the latest comment): `screenshot_E.png`
- Python source code: `code_E.py`

(d) Verification of the alternative explanation (5pts) **[Return before 11am45]**

Plot the ticket fare histograms again, but now separately for the male and female passengers. Does the ticket fare explain who survived or the gender? (you may find the `matplotlib.pyplot.subplot()` useful)

Submitted items:

- Full desktop screenshot of running your code and printing the two histograms (must include Slack with the latest comment): `screenshot_F.png`
- Python source code: `code_F.py`

Note that we did not study the effect of age at all. You may continue your analysis after the test. Who really survived from the terrible accident, and what could explain that?